Pranay Mundra

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Education

University of California, Santa Cruz – *Master of Science in Computer Science* GPA - 3.89
Advanced Computer Architecture, Distributed Systems, Advanced OS, Algorithms Sept 2024 – Aug 2026

PES University, Bengaluru – *B.Tech in Electronics & Communication* CGPA - 9.01
Control System Theory, Digital Communication, Robotics and TA for *UE19EC339* Aug 2018 – Aug 2022

Experience

University of California, Santa Cruz | *Graduate Teaching Assistant* Jan 2025 – Jun 2026

- I mentor students in *CSE115A/B/C* working on software projects sponsored by companies such as Nutanix and Keysight Technologies. My responsibilities include requirements analysis, high-level system design and sprint planning.

MakeMyTrip & Goibibo (NASDAQ: MMYT) | *Software Engineer I* Jun 2022 – Jun 2024

- Part of India's largest online travel agency overseeing A/B infrastructure, Discounting and Persuasion engines.
- Designed a highly concurrent multi-org *ConfigKey* store supporting the A/B infrastructure using a two level caching architecture with Aerospike giving sub-millisecond p99 latencies at 300K requests per second, down from 50ms.
- Implemented a rule based rate limiter to optimize notification delivery across SMS, WhatsApp, and email, thereby increasing relevant and timely user engagement while also reducing infra costs by upto 40%.
- Revamped loyalty programs (MMTBLACK, GoTribe, myPartner Hero) with configurable aggregations per user, thus supporting newer programs that can target any type of customer.
- Fixed widespread fraud in Refer&Earn (upto $50K) by adding constraints on mobile number within the sign up flow.
- Added virtual thread support to the legacy A/B platform in Java 8 using `quasar` thus reducing CPU footprint by upto 40%

Software Intern Jan 2022 – May 2022

- Setup cache sync for intra-cluster services using Kafka and parallelized `/assignVariant` of the A/B platform.
- Support for the legacy route planning service before it was enhanced for How2Go.

ARTPARK @ Indian Institute of Science | *Research Intern* Jun 2021 – Dec 2021

- Built a log aggregator with Elastic that informs warehouse robot operators of any anomalies in the environment.
- Designed a ROS2 safety node integrated with warehouse cameras to stop a robot based on human proximity.

Selected Projects

Atropos | *Go, OpenTelemetry, gRPC, Kubernetes, k6, Vegeta* github

- Instrumentation SDK that embeds at service boundaries (HTTP middleware, gRPC interceptors) to inject faults (latency, TCP toxics, CPU/IO stress) and freeze services into cached-response replay via a *cache-box* primitive for isolating per-service contributions to tail latency under concurrent load.
- Built zeus-go, a load orchestration platform with declarative JSON workflow DAGs, persona-based k6 sidecars, and a policy engine that auto-triggers Vegeta attacks; W3C Baggage propagation for end-to-end trace correlation.

Global Upgrade Safety & Maximal Safe Subset | *Go, OpenAPI, Protobuf* github

- Static analysis tool that decides whether simultaneous schema changes across a microservice mesh can co-deploy safely; when they cannot, computes the MSS in linear time (reduction to `MAX-TRUE` Horn-SAT) and pinpoints the fields and type mismatches responsible.
- Four-schema edge model (Send/Accept/Return/Expect) with endpoint-only data-flow annotations and automatic chain discovery for detecting multi-hop integrity violations invisible to pairwise tools like Buf and Pact.

Slug MCP | *Rust, rmcp, Moka cache* github

- Model Context Protocol server that exposes UCSC campus services (dining menus and nutrition, live gym occupancy, study room booking, class search with enrollment data, transit predictions) as structured tool calls for LLM integration over stdio/SSE transport. Add the connector - `https://2262-cse115b-02.be.ucsc.edu/mcp`
- Instruments Santa Cruz County with environmental and research telemetry (surf and wave buoys, weather and tide forecasts, air quality, wildfire detections, ocean monitoring with harmful-algal-bloom risk and coastal upwelling indices, citizen-science wildlife observations) as MCP tool calls for UCSC researchers and LLM workflows.

Technical Skills

Languages: Go, Rust, Java, Python, C/C++, SQL, JavaScript, Shell
Frameworks & Libraries: OpenTelemetry, gRPC, vert.X, Spring, gin, k6, Vegeta
DevOps & Tools: Docker, Kubernetes, Jaeger, Prometheus, Grafana, Jenkins, AWS, git
Databases & Queues: MongoDB, MySQL, Aerospike, Redis, Kafka, RabbitMQ
Areas of Expertise: distributed systems, fault injection, tracing & observability, SDK design, scalable web services